

Table 8: Answer table for Q.9 over Table 3 of Ghazy et al. [25], modeled in the ORKG as Comparison R1471077. For each ligand family, the table lists representative materials, GPC and temperature ranges, and co-reactants for high-GPC (≥ 1 Å) and low-temperature (≤ 250 °C) rare-earth ALD processes.

Ligand family	Representative materials	GPC [Å]	Temp [°C]	Co-reactant
Cp-derived	Y ₂ O ₃ , Er ₂ O ₃ , Gd ₂ O ₃ , Tm ₂ O ₃ , GdAlO ₃	1.0–4.5	175–250	H ₂ O, O ₃ , TMA
Amidinate / Formamidine	La ₂ O ₃ , CeO ₂ , Pr ₂ O ₃ , Gd ₂ O ₃	1.1–2.8	200–240	H ₂ O, O ₃
Guanidinate	CeO ₂ , Y ₂ O ₃ , Dy ₂ O ₃ , Gd ₂ O ₃	1.0–2.1	150–200	H ₂ O
β -Diketionate	LaF ₃ , YF ₃ , HoF ₃	1.0–2.0	200–225	TiF ₄ , NbF ₅

groups the resulting rare-earth ALD processes by ligand family and summarizes representative materials, GPC ranges, temperature ranges, and co-reactants.

Table 9: Answer table for Q.10 over Table 3 of Ghazy et al. [25], modeled in the ORKG as Comparison R1471077. For Y₂O₃ processes in the 200–300 °C window, the table reports ALD mode, average GPC, record counts, and example co-reactants.

Mode	Avg. GPC [Å]	Records	Example co-reactant
Thermal ALD	1.95	28	Y(thd) ₃
PE-ALD	1.65	3	O ₂ plasma

Complex Query. The tenth question is: “For Y₂O₃ only, compare the average growth per cycle (GPC) between plasma-enhanced ALD (PE-ALD) and thermal ALD within the 200–300 °C window. Return mode (PE vs. thermal), average GPC, count of records, and a sample list of co-reactants.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/alld-y2o3-gpc200> over the ORKG comparison R1471077 of Table 3 in the review article. The tabulated answer is in Table 9, which summarizes the average GPC, record counts, and example co-reactants for thermal and PE-ALD Y₂O₃ processes in this temperature range.

3.3.3 Machine-actionable modeling of Table 4

We modeled Table 4 of Ghazy *et al.* (2025) as an ORKG comparison available at <https://orkg.org/comparisons/R1470110> [3]. The comparison records, for each reported ALD/MLD process, the rare-earth–organic hybrid material, metal precursor (e.g., R(thd)₃, R(dpdmg)₃), organic linker (e.g., TPA, PDC, NDC), growth per cycle (GPC), deposition temperature, and literature reference.

The table thus covers classical R(thd)₃ routes with aromatic linkers as well as newer guanidinate-based chemistries reporting higher GPC. In the ORKG, this structured representation forms the basis for the Easy and Complex Queries on rare-earth–organic hybrid films introduced next.

Table 10: Answer table for Q.11 over Table 4 of Ghazy et al. [25], modeled in the ORKG as Comparison R1470110. The table lists rare-earth ALD/MLD hybrid films with GPC ≥ 5 Å at deposition temperatures ≤ 250 °C, including material system, metal precursor, organic precursor, GPC, and temperature.

Material system	Metal precursor	Organic precursor	GPC [Å]	T [°C]
Ce-TPA	Ce(DPDMG) ₃	TPA	5.4	200
Er-3,5PDC	Er(DPDMG) ₃	3,5PDC	6.4	245
Eu-HQA	Eu(thd) ₃	HQA	7.3	210

Easy Query. The eleventh question is: “List all rare-earth ALD/MLD hybrid films that achieve high growth per cycle (GPC ≥ 5 Å) at low deposition temperature (≤ 250 °C). Report the material system, metal precursor family, organic precursor, GPC, and temperature; sort by GPC.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/HiGLoT-query>

over the ORKG comparison R1470110 of Table 4 in the review article. The tabulated answer is in Table 10, which lists the corresponding high-GPC, low-temperature hybrid film processes and their deposition conditions.

Complex Query. The twelfth question is: “Group rare-earth ALD/MLD hybrid films by organic linker family (terephthalate, pyridinedicarboxylate, naphthalenedicarboxylate, pyrazine-based, other) and compute the average growth per cycle (GPC) for each, considering only films deposited at temperatures ≤ 250 °C. Report linker family, average GPC, and number of films; sort by average GPC.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/almdl-linker-gpc> over the ORKG comparison R1470110 of Table 4 in the review article.

3.3.4 Machine-actionable modeling of Table 5

We modeled Table 5 of Ghazy *et al.* (2025) as an ORKG comparison available at <https://orkg.org/comparisons/R1469991> [24], instantiated with the template <https://orkg.org/templates/R1469957>. The comparison records, for each Er^{3+} -doped MOSLED device, the host matrix, annealing temperature, external quantum efficiency (EQE), power efficiency (PE), threshold voltage, emission lifetime (τ), and operational device lifetime (OLT). This machine-actionable representation provides a machine-actionable basis for queries on how host composition and processing conditions relate to device performance in Er^{3+} -doped MOSLEDs.

Table 11: Answer table for Q.13 over Table 5 of Ghazy et al. [25], modeled in the ORKG as Comparison R1469991. The table lists Er^{3+} -based MOSLED host matrices with $\text{EQE} \geq 10\%$ and a reported threshold voltage, together with EQE, voltage, annealing temperature, emission lifetime (τ), and operational lifetime (OLT).

Host matrix	EQE [%]	Voltage [V]	Anneal [°C]	Lifetime τ [ms]	OLT [h]
$\text{Al}_2\text{O}_3:\text{Yb}$	24	14	—	0.9	—
Ga_2O_3	36	15	900	2.04	100
$\text{Lu}_3\text{Al}_5\text{O}_{12}$	10	50	1100	1.65	—

Easy Query. The thirteenth question is: “Among Er^{3+} MOSLEDs in the comparison, which host matrices achieved high external quantum efficiency ($\text{EQE} \geq 10\%$) at the lowest threshold voltage, and what annealing temperatures and lifetimes (τ , OLT) were reported?” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/mosled-high-eqe> over the ORKG comparison R1469991 of Table 5 in the review article. The tabulated answer is in Table 11, which lists the corresponding host matrices together with EQE, threshold voltage, annealing temperature, emission lifetime, and operational lifetime.

Table 12: Answer table for Q.14 over Table 5 of Ghazy et al. [25], modeled in the ORKG as Comparison R1469991. For each Er^{3+} -based MOSLED host matrix, the table lists EQE, operating voltage, the derived EQE/Vol metric, annealing temperature, emission lifetime (τ), and operational lifetime (OLT).

Host matrix	EQE [%]	Vol [V]	EQE/Vol	Ann. T [°C]	τ [ms]	OLT [h]
Ga_2O_3	36.0	15	2.40	900	2.04	100
$\text{Al}_2\text{O}_3:\text{Yb}$	24.0	14	1.71	—	0.90	—
$\text{Yb}_3\text{Al}_5\text{O}_{12}$	5.2	5	1.04	1100	1.20	—
$\text{Lu}_3\text{Al}_5\text{O}_{12}$	10.0	50	0.20	1100	1.65	—
Yb_2O_3	8.5	60	0.14	1000	—	—
$\text{Y}_3\text{Ga}_5\text{O}_{12}$	2.5	25	0.10	800	2.26	49
$\text{Gd}_3\text{Ga}_5\text{O}_{12}$	1.9	30	0.06	1000	1.59	—
ZnGa_2O_4	0.3	30	0.01	800	0.82	5.5

Complex Query. The fourteenth question is: “Compute an efficiency-per-volt metric (EQE/Vol) to approximate the EQE-voltage trade-off, and rank host matrices accordingly.” The SPARQL query

implementation for this question can be rerun via <https://tinyurl.com/mosled-pareto-score> over the ORKG comparison R1469991 of Table 5 in the review article. The tabulated answer is in Table 12, which lists, for each host matrix, EQE, operating voltage, EQE/Vol, annealing temperature, emission lifetime, and operational lifetime.

3.3.5 Cross querying across the modeled tables

Machine-actionable modeling of the review tables enables not only within-table queries but also cross-table queries that combine complementary information into a unified analytical view. This allows users to derive aggregate, precise insights that cannot be obtained from any table in isolation. In this section, we illustrate how Tables 2, 3, and 5 of Ghazy *et al.* (2025) can be jointly queried: Table 2 provides host materials that are Er-doped or luminescent, Table 3 supplies their corresponding ALD process recipes, and Table 5 reports MOSLED device metrics for Er-doped systems. Table 4, which focuses on rare-earth-organic hybrid films, has limited overlap with the MOSLED data in Table 5 and is therefore treated as optional and not included in the examples that follow.

Table 13: Answer table for Q.15 joining Tables 3 and 5 of Ghazy et al. [25], modeled in the ORKG as Comparisons R1471077 (Table 3) and R1469991 (Table 5). For each Er^{3+} -based MOSLED host matrix, the table lists the corresponding ALD process with the lowest reported deposition temperature, including process material, metal precursor, co-reactant, precursor family, GPC, and deposition temperature, together with EQE.

Host matrix	EQE [%]	Process material	Metal precursor	Co-reactant	Family	GPC [$\text{\AA}/\text{cycle}$]	T [$^{\circ}\text{C}$]
$\text{Gd}_3\text{Ga}_5\text{O}_{12}$	1.9	$\text{Gd}_3\text{Ga}_5\text{O}_{12}$	$\text{Gd}(\text{thd})_3$	TEGa	R(thd)x	0.19	350
$\text{Lu}_3\text{Al}_5\text{O}_{12}$	10.0	$\text{Lu}_3\text{Al}_5\text{O}_{12}$	$\text{Lu}(\text{thd})_3$	TMA	R(thd)x	0.19	350
$\text{Y}_3\text{Ga}_5\text{O}_{12}$	2.5	$\text{Y}_3\text{Ga}_5\text{O}_{12}$	$\text{Y}(\text{thd})_3$	Et_3Ga	R(thd)x	0.23	350
Yb_2O_3	8.5	Yb_2O_3	$\text{Yb}(\text{MeCp})_3$	H_2O	other	1.00	180
$\text{Yb}_3\text{Al}_5\text{O}_{12}$	5.2	$\text{Yb}_3\text{Al}_5\text{O}_{12}$	$\text{Yb}(\text{thd})_3$	TMA	R(thd)x	0.20	350

Easy Query 1. The fifteenth question is: “Show the ALD recipe alongside device performance for Er-MOSLED hosts.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t3t5-lowT> over the ORKG comparisons R1471077 (Table 3) and R1469991 (Table 5). For each Er^{3+} -based MOSLED host matrix, it retrieves the highest reported EQE in Table 5 and joins it with the corresponding ALD process parameters from Table 3 (process material, metal precursor, co-reactant, precursor family, GPC, and deposition temperature). The tabulated answer is in Table 13.

Table 14: Answer table for Q.16 joining Tables 2 and 5 of Ghazy et al. [25], modeled in the ORKG as Comparisons R1469955 (Table 2) and R1469991 (Table 5). For each luminescent rare-earth-doped material appearing in both tables, the table lists EQE, threshold voltage, power efficiency (PE), annealing temperature, emission lifetime (τ), and operational lifetime (OLT).

Material	EQE [%]	Threshold [V]	PE [$\times 10^{-4}$]	Ann. [$^{\circ}\text{C}$]	T	τ [ms]	OLT [h]
Ga_2O_3	36	~ 15	81	900	2.04	100	
$\text{Gd}_3\text{Ga}_5\text{O}_{12}$	1.9	30	3.6	1000	1.59	—	
$\text{Lu}_3\text{Al}_5\text{O}_{12}$	10	50	12	1100	1.65	—	
$\text{Y}_3\text{Ga}_5\text{O}_{12}$	2.5	25	3.8	800	2.26	49	
Yb_2O_3	8.5	60	10	1000	—	—	
$\text{Yb}_3\text{Al}_5\text{O}_{12}$	5.2	~ 5	4.8	1100	1.20	—	
ZnGa_2O_4	0.3	30	—	800	0.82	5.5	

Easy Query 2. The sixteenth question is: “Which luminescent materials listed as doped systems in the ALD dopant overview also have measured MOSLED performance reported in the rare-earth MOSLED performance comparison?” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/LumMOSLED-query> over the ORKG comparisons R1469955 (Table 2) and R1469991 (Table 5). For each luminescent host material that appears in both tables, it retrieves the measured performance parameters—EQE, threshold voltage, power efficiency (PE),